WEEK 6 / LESSON 4 / 50 MINUTES

L4

Watch the model age

A fraud-detection model ran for 14 months without monitoring. Recall drifted from 0.89 to 0.52 as fraudsters adapted. The company discovered it from an audit — not an alert. Drift kills silently.

Which of your models is silently drifting right now?

WHAT'S INSIDE

WHY THIS LESSON MATTERS

₹18 crore lost. 14 months undetected. One missing dashboard.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1. Data drift vs concept drift.
2. PSI + KS for drift detection.
3. Prometheus + Grafana model dashboards.
4. Alerts that trigger retraining.
5. Drift pipeline for your model.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01
Data · Label · Concept drift
Three kinds; different causes

02
PSI + KS — the practical...
Key concept of this lesson.

03
The 7 dashboards per model
What to watch

04
Alerts that trigger action
Noise vs signal

01

PART 1 OF 4

Data · Label · Concept drift

Three kinds; different causes

CONCEPT 1 OF 4

Data · Label · Concept drift

Data drift — P(X) shifts (new customer mix).
Label drift — P(y) shifts (fraud rate falls).
Concept drift — P(y|X) shifts (mapping changes).
Data drift: easiest to detect. Concept drift: hardest — needs prod labels.
3 KINDS
Different symptoms

All three monitored weekly.

DEFINITION

Data · Label · Concept drift

Three kinds; different causes

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

PSI + KS — the practical tests

A core idea you'll use throughout this lesson.

KEY POINTS

- A core idea you'll use throughout this lesson
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

CONCEPT 3 OF 4

The 7 dashboards per model

1. Prediction distribution.
2. Feature drift (PSI per feature).
3. Model performance on labelled prod.
4. Latency p50, p95, p99.
5. Throughput (QPS).
6. Error rate.
7. Business metric (revenue, conversion).

COVERAGE
7 dashboards

Less than 7 = blind spots.

DEFINITION

The 7 dashboards per model

What to watch

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Alerts that trigger action

Data drift: PSI > 0.25 on ≥ 3 features → warn.
Performance: drop > 5% vs baseline → page on-call.
Two alerts in a week → auto-trigger retraining.
Alerts without action = noise. Retire them.
RULE
Alert → Action

Or don't alert.

KEY POINTS

- Data drift: PSI > 0.25 on ≥ 3 features → warn
- Performance: drop > 5% vs baseline → page on-call
- Two alerts in a week → auto-trigger retraining
- Alerts without action = noise. Retire them

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK · PAIR · SHARE

ACTIVITY

Pause and reflect

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4 Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1 *Where will you apply this in your next project?*

Q2 *What was the single biggest 'aha' from this lesson?*

Q3 *Which habit will you start using this week?*

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

KNOWLEDGE CHECK · QUESTION 1 OF 3

PSI threshold for severe drift:

A > 0.1

B > 0.25

C > 0.5

D > 1

ANSWER · QUESTION 1 OF 3

PSI threshold for severe drift:

CORRECT ANSWER

B

> 0.25

WHY

>0.25 = severe

KNOWLEDGE CHECK · QUESTION 2 OF 3

Hardest drift to detect:

A Data

B Label

C Concept

D All equal

ANSWER · QUESTION 2 OF 3

Hardest drift to detect:

CORRECT ANSWER

C

Concept

WHY

Concept drift — needs fresh labels

KNOWLEDGE CHECK · QUESTION 3 OF 3

7 dashboards — one NOT on the list:

- **A** Latency
- **B** Prediction dist
- **C** CPU usage
- **D** Business metric

ANSWER · QUESTION 3 OF 3

7 dashboards — one NOT on the list:

CORRECT ANSWER

C

CPU usage

WHY

CPU is infra; not a model-specific dashboard

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Data drift — P(X) shifts (new customer mix).
Label drift — P(y) shifts (fraud rate falls).
Concept drift — P(y|X) shifts (mapping changes).
Data drift: easiest to detect. Concept drift: hardest — needs prod labels.
3 KINDS
Different symptoms

All three monitored weekly.

THE TAKEAWAY

The right tool, applied to the right problem, beats the loudest idea every time.

WHAT TO NOTICE

1. Data · Label · Concept drift
2. PSI + KS — the practical…
3. The 7 dashboards per model
4. Alerts that trigger action

APPLY IT → ASSIGNMENT

Every model decays.

"Plan for it before it happens — not after."

DELIVERABLES

- NEXT
- Lesson 5 — Feature stores.
- Document one decision you made because of this lesson.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

ONE THING TO REMEMBER

“

Knowing the right tool is half the battle. Doing the small thing today is the other half.

— AI Learning Hub

END OF LESSON 6

What you can now do

Each one of these is now part of your working vocabulary.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.